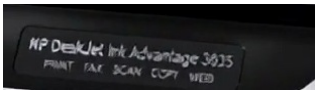
Multimodal Judges Flag, Humans Decide: Supplementary Material

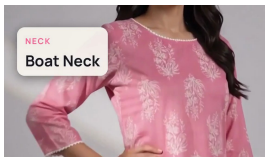
A. What the Review Catches

Table S1 shows real ads with every axis the review asks about (Tab. 1) and every kind of problem but the music: frames from the 120-ad round and, where the round has no clear case, from production runs or our studies. A product defect is shown beside the product's catalog photo. The Fix column gives the fix for the problem (Tab. 3); where the scene was actually fixed, it shows the result. Each row says who caught the problem; in the round reviewers saw the judges' findings, so a judge flag and the marks of reviewers who accepted it count as one agreement. The defects shown come from our generation pipeline and are not in the real products; no brand shown is affiliated with this work or endorses it. Faces are blurred. The icon at the end of a row opens the example on our public gallery page (▶ with its clip, 🔊 with its audio, 🖼 with its images only).

Table S1. What the review catches: one example of each kind of problem but the music, with the fix.

Problem	What the review saw	Fix	What is wrong, and who caught it
Product			
Wrong product, replaced by a still	 catalog photo  before, 0:03	 after, 0:04 *Replaced by an animated still*	Eyeliner ad from a production run. The applicator is drawn as a kinked, bristly brush, where the real one has a smooth precision stem (0:02-0:05); judge flagged, 1 of 1 reviewer marked. After: the scene was replaced by an animated still of the product photo (a slow pan, no generated motion), which shows a straight stem; not judged. 🖼
Shape wrong	 catalog photo (crop)  frame 0:08	Borrow, generate or edit the scene	Beard trimmer. The ridged length-adjustment wheel is replaced by an invented see-through battery chamber (0:07-0:11). Judge flagged, 3 of 3 reviewers marked. ▶
Color wrong	 catalog photo  frame 0:00	Borrow, generate or edit the scene	Smartwatch. The strap buckle is bright silver, where the real watch's is dark blue (0:00-0:01); judge flagged, 1 of 1 reviewer marked. ▶
Text slightly garbled (low), Repair at higher resolution	 catalog photo  before, 0:05 (crop)	 after, Repair (crop) *Repair the scene in place*	Inkjet printer. Before: the model name and the small print under it, PRINT FAX SCAN COPY WEB in the catalog photo, are smeared but readable (0:05); judge flagged, 1 of 1 reviewer marked. After: Repair re-rendered the scene from its original inputs at higher resolution, and both lines read correctly; this pair is from our repair study and is judged here by eye only. 🖼
Text gibberish, color drift	 catalog photo  frame 0:04 (white body; number blurred)  frame 0:07, service badge (crop; number blurred)	Borrow, generate or edit the scene	Water purifier. The service badge turns into non-words with a wrong phone number, and the body starts white before turning black like the real unit (0:04-0:09). Judge flagged both, 3 of 3 reviewers marked both. ▶

Problem	What the review saw	Fix	What is wrong, and who caught it
Garbled label, Repair at higher resolution	 catalog photo (crop)  before, 0:11 (crop)	 after, Repair (crop) *Repair the scene in place*	Body lotion. Before: the claim lines on the lower label are smeared (0:11); judge flagged, 3 of 3 reviewers marked. After: Repair regenerated the clip itself at higher resolution, guided by its own prompt and references, and the lines read correctly; graded by four blind model graders only.
AI issue on the product	 catalog photo (crop)  frame 0:12	Borrow, generate or edit the scene	Suitcase, last scene (0:09-0:15). Two pairs of spinner wheels stick out of the case right beside the combination lock; on the real case the lock sits high on the side seam and the wheels are only at the four bottom corners. The scene judge flagged it as an AI issue but the product judge passed it; 1 of 1 reviewer marked the product wrong in an early pilot round.
Scene quality			
Good scene (Keep)	 frame 0:06.3 (start, wide)  frame 0:08.6 (end of the push-in)	Keep the scene	Kurta set, a good scene for contrast: the camera starts wide on the outfit and pushes in on the print and hem while a caption names the cut (0:06-0:09). The reviewer guide uses it as a Keep example (1 pilot reader marked Keep); the scene judge raised no finding on it.
Ghost copies overlap	 frame 0:08.4 (just before)  frame 0:09.0 (copies overlap)	Borrow, generate or edit the scene	Wristwatch, strap close-up. Half a second into the scene, as the camera changes angle, several translucent copies of the watch face and buckle overlap, then settle into one strap (about 0:09). Judge flagged (frames judge accepted 3/3, clip judge 2/3); 3 of 3 reviewers marked swap (AI issue).
Crowded frame	 frame 0:03.4  frame 0:05.6	Borrow, generate or edit the scene	Air conditioner. A cluster of duplicated indoor units and an oversized copper coil fill the frame, and then the coil fills most of the frame, so the product is hard to read (0:03-0:07). Judge flagged (accepted 3/3); 3 of 3 reviewers marked swap (repetitive, dull or crowded).
Repeats the opening	 opening scene: frames 0:00.2 and 0:03.7  closing scene: frames 0:11.7 and 0:15.0	Borrow, generate or edit the scene	Backpack. The closing scene repeats the opening scene: the same front-on framing and the same push-in to the printed panel (0:00-0:04 and 0:11-0:15). Both scene judges flagged it (each accepted 3/3); 3 of 3 reviewers marked the closing scene swap (repetitive, dull or crowded).
Set changes mid-ad	 frame 0:02.3 (scene 2)  frame 0:10.2 (scene 4)	Borrow, generate or edit the scene	Suitcase ad from a production run. Scenes 1 to 3 sit on a cream studio background; at 0:09.4 scene 4 cuts to a dark set, and the last scene goes back to cream. Judge flagged (accepted 1/1); 1 of 1 reviewer marked swap (world or continuity breaks).
Units multiply	   frame 0:12.6 frame 0:12.8 frame 0:15.0 (end)	Borrow, generate or edit the scene	Air conditioner, closing scene. One indoor unit becomes four, plus an outdoor unit and a copper coil: they fade in from nowhere and drift into place (0:12.4-0:15). Clip judge flagged (accepted 3/3; frames judge 2/3); 3 of 3 reviewers marked swap (impossible motion).

Problem	What the review saw	Fix	What is wrong, and who caught it
Shoe floats through model	 frame 0:00.5 frame 0:00.9 frame 0:01.4	Borrow, generate or edit the scene	Sports shoe, opening scene. A giant translucent shoe fades in over the model and ends up floating in front of him, passing through his body (0:00.5-0:01.5). Both scene judges flagged it (each accepted 3/3); 3 of 3 reviewers marked swap (impossible motion).
Ends on a tight crop	 frame 0:13.0 frame 0:14.9 (last frame)	Borrow, generate or edit the scene	Water purifier, closing scene. The scene opens on the front of the purifier, then pushes in until the ad ends on a tight crop of the tap (last frame 0:14.9). Both scene judges flagged it (each accepted 3/3); 3 of 3 reviewers marked swap (weak ending).
Pronunciation			
Word said wrong	 frame, scene 2 (caption card) PRONUNCIATION · SAID VS HEARD ON SCREEN Boat Neck NARRATION "Bolt neck" *A product attribute turned into another word. Audio (3 s) on the gallery page.*	Respell the word and re-voice the line	Kurta set, scene 2 of 4: the on-screen card reads 'Boat Neck', but the narrator says 'bolt neck'. 1 reviewer marked it in an early pilot round; no judge result is recorded for it.
Respelled, re-voiced	PRONUNCIATION · RESPELL THE WORD, RE-VOICE THE LINE SCRIPT LINE Straight fit, ready to wear. BEFORE: HEARD "Street fit, ready to wear." FIX Voice prompt told to say "Straight" as STRAYT; the written line is unchanged. Only this line is re-voiced. AFTER: HEARD "Straight fit, ready to wear." CONTROL Plain second take, no respelling (another line): "Mate from …" became "Made from ridiculously soft cotton." *"Heard" is the speech judge's transcript; no person rated these takes. Audio on the gallery page.*	*Word respelled, line re-voiced*	Trousers ad, line 'Straight fit, ready to wear.': the speech judge heard 'Street fit'. After the word was respelled STRAYT in the voice prompt and only that line was re-voiced, it heard 'Straight fit'. Only the speech judge checked the two takes; no person did. Control: a plain second take with no respelling turned 'Mate from' into 'Made from' on another line, and across the whole test respelling fixed about as many flagged lines as a plain second take (27% vs 25%).

Problem	What the review saw	Fix	What is wrong, and who caught it
Script			
Claim not in the source	 frame 0:01.5 SCRIPT · UNSUPPORTED CLAIM NARRATION 0:00 "Step into travels that demand **unbreakable**, lasting resilience." SCRIPT JUDGE "source says shell is 'scratch-resistant' and 'tough' but never asserts it is 'unbreakable', an absolute the weak marketing sentence does not support." *The on-screen card repeats the claim ("Unbreakable Resilience").*	Correct the claim or regenerate the line	Suitcase ad, 0:00-0:04: the narration says the shell is 'unbreakable', but the product record only calls it scratch-resistant and tough. The script judge flagged it, and all 3 reviewers marked it by accepting that finding, so the marks are not independent of the judge.
Number without its unit	 frame 0:07 SCRIPT · AWKWARD PHRASING NARRATION 0:06 "Power through with a **5000** battery." SCRIPT JUDGE "'5000' is a numeral stripped of its unit (mAh), reading like a spec field value rather than natural speech."	Rephrase or regenerate the line	Smartphone ad from a production review outside the round, 0:06: the line 'Power through with a 5000 battery.' drops the unit (mAh), so it sounds like a spec-sheet field. The script judge flagged it, and so did a second script judge that this review also ran and that was later dropped (each finding got 1 thumbs-up and none down). Both reviewers marked awkward phrasing: one on this line, one for the ad as a whole.
Claim narrowed	SCRIPT · CLAIM REWRITTEN AT GENERATION (TEST) BEFORE "The [brand] **block it all** with intelligent AI noise cancellation." LISTING blocks out background noise on calls; filters out background noise AFTER "The [brand] **block background noise on calls**." RESULT Only this line was rewritten. The script judge passed it on its first re-read; a separate model check agreed.	*Line regenerated at generation*	Earbuds ad from the 410-ad script-rewrite test of Sec. 5: 'The [brand] block it all with intelligent AI noise cancellation.' claims more than the listing, which only says they block background noise on calls and filter out background noise. Only that line was rewritten at generation time, as 'The [brand] block background noise on calls.' The script judge flagged the old line and passed the new one on its first re-read, and a separate model check agreed. The script is one of the 129 the annotators then read cold, and no annotator flagged it.

Problem	What the review saw	Fix	What is wrong, and who caught it
Sound			
Wrong accent	SOUND · THE VOICE CLIP: Whole narration, 0:00–0:13, skincare lotion ad PROBLEM: Meant for the Indian market, but the narrator did not have the right accent CAUGHT BY: Sound judge flagged it; 2 of 2 reviewers marked the whole voice off *Audio (13 s) on the gallery page.*	Re-voice the ad	Skincare lotion ad from a production run before the round, whole narration (0:00-0:13): the ad was meant for the Indian market, but the narrator did not have the right accent. The sound judge flagged it (in the round it flagged no ad, Sec. 6), and both reviewers marked the whole voice as off after accepting its finding.
Voice swapped	SOUND · VOICE SWAPPED (REVIEW EDITOR QUICK FIX) BEFORE: First stock narrator. The run's one reviewer marked the whole voice off. FIX: Quick fix: narrator swapped to a different stock voice. Same script, same music. AFTER: Re-rendered with the new voice. Not judged by a model or a person. *The after render also carries an earlier scene-2 remake, so compare the audio only. Audio (8 s before, then 8 s after) on the gallery page.*	*Voice swapped*	Eyeliner ad from a production run, whole narration: the run's one reviewer marked the whole voice off, and a Quick fix in the review editor swapped in a different stock voice, keeping the same script and music. This is a plain voice swap, separate from the speech-to-speech re-voicing of Tab. 3. No judge flagged the voice, and no model or person judged the new render. The new render also has an earlier scene-2 remake, so compare the audio only.
Captions			
Card over the wheel	 frame 0:02	Move or restyle the caption	Suitcase ad, 0:00-0:04: the caption sits on the wheel and bottom corner of the suitcase even though the frame has empty space around it. 3 of 3 reviewers marked placement; a placement judge, run after the round and never shown to reviewers, flagged it in both of its runs.
Card over a face	 frame 0:11 (face blurred)	Move or restyle the caption	Face serum ad, 0:10-0:12: the callout card sits on the model's forehead (face blurred here, card left as rendered). 2 of 3 reviewers marked placement; the placement judge, run after the round and never shown to reviewers, flagged it in both of its runs.
Caption about a hidden part	 frame 0:13	Reword or remove the caption	Sports shoe ad, 0:12-0:15: the card reads 'Outsole: Reliable traction', but the shoes stand flat and the sole's tread is never shown. 2 of 3 reviewers marked relevance; the relevance judge, run after the round and never shown to reviewers, flagged it in both of its runs.
Overlay restyled (unshipped prototype)	 before: as delivered, 0:12	 after: prototype restyle (never shipped) *Caption restyled (prototype)*	Router ad, 0:12: the delivered caption is near-black text on a navy background and cannot be read. Restyling only the caption layer put the same words on a cream card without touching the video. This is an unshipped prototype, and only the project owner looked at the after. Reviewers are not asked about readability, but 2 of 3 marked this caption's placement; the placement judge passed it in both runs.

B. The Same Ads with Judges Hidden

R1, a reviewer who had not seen the ads, reviewed the round's ads in a separate run of the portal with every judge hidden. The run held 60 of the 120 ads, three per product, in an order that cycles through the 20 products; every question but sound was required and the secondary options were offered (a blank sound answer counts as no issue; sound was left blank on every ad). The units, the rules that turn answers into marks (those of the round), the outcomes, the bootstrap and the exclusions were written down and the analysis code hashed before the first review; one amendment, made before any review was submitted, records the scene-quality confound below. R1 submitted all 60 ads. Checks: every judge was hidden, the run holds no thumbs, and the stored findings on the 60 ads are those of the round (360 of 360 ad–judge pairs unchanged). Pronunciation was computed but is not reported, because the narration engine and its judge changed after the round.

Table S2 compares, on the same ads and units, each reviewer alone, the stored judge findings and the round's reviewers, who saw them. A unit is marked as in the round: a scene with the product marked wrong or a swap the reviewer chose, a line marked bad, a caption marked for placement or relevance. Intervals are percentiles of 2,000 bootstrap resamples of ads. †In the round, marking a scene's product wrong pre-filled a swap for it, and an older grid hid such scenes from the scene-quality question, so the round was asked about fewer scenes (in parentheses) and its scene-quality marks mix the reviewer, the findings and the pre-fill.

Time is descriptive, since the two groups have different reviewers. Per-question gaps over 15 minutes count as breaks and are dropped; per-review times run from the first answer to submission, dropping reviews over 4 hours (one of R1's). R1 took a median of 3.4 [2.4, 4.7] minutes from first answer to submission (quartiles in brackets; 59 reviews), against 4.1 [2.8, 6.6] for the round's 139 reviews of the same ads. Median time per question, R1 against the round: scene quality 44 s against 0 s; script 43 s against 65 s; caption placement 30 s against 7 s; caption relevance 1 s against 2 s. The product is answered first, and the review record starts at the first answer, so its time cannot be measured; in the round the product answer pre-filled the scene-quality grid, so its scene-quality time is near zero.

R2, one of the round's reviewers, later reviewed all 60 ads in the same run with every judge hidden, under a second amendment written before any of these reviews was read. R2 had also reviewed the same 60 ads in the round about two weeks earlier, with judge findings on screen, and the amendment pre-specified the comparison of the two passes. On the same scenes, the product was marked wrong on 57% [49, 65] with judge findings and 19% [13, 26] with judges hidden (κ 0.22 between the passes); of the 7 lines the script judge flagged, 7 were marked with its findings on screen and 1 without. The pass with judges hidden came second, so memory of the findings would, if anything, narrow these gaps. A review by R2 took a median of 2.3 [1.5, 4.4] minutes with judges hidden (59 timed reviews; quartiles in brackets), against 3.0 [2.3, 4.2] with judge findings in the round; this was R2's second look at these ads, so the shorter time cannot be credited to hiding the judges. Pooled over both reviewers, a review with judges hidden took a median of 2.9 minutes (118 timed reviews). The two reviewers working alone agreed, on the 60 ads both reviewed, at Cohen's κ product 0.17 [0.02, 0.31]; scene quality 0.14 [0.04, 0.24]; script -0.04 [-0.05, -0.02]; caption placement 0.06 [-0.11, 0.25]; caption relevance -0.02 [-0.03, 0.00].

Over the session. Section 4 gives R2's product marks over the first, middle and last 20 ads by submission time, with judges hidden and with judge findings, and R2's review times with judges hidden (Fig. 5b); on the same ads the round's majority marked 56%, 54% and 62%. Against the round's majority on each ad, the fall in R2's marks with judges hidden has Spearman ρ -0.27 over submission order (permutation p 0.04); total marks per ad have ρ -0.56 and minutes per review -0.59 (both $p < 0.001$). R1's product marks held (31%, 37% and 30%; ρ -0.06, p 0.65) while reviews sped up from 4.3 to 2.5 minutes (ρ -0.45, $p < 0.001$). This analysis was added after the results were read.

C. Baselines in Detail

Table S3 lists the primary score of every product-fidelity baseline of Tab. 4 on the 120 blind-labeled scenes (29 defective, 35 ads), with 95% intervals from the same ad-clustered bootstrap (2,000 resamples) and the paired gap to the deployed clip judge. Lower predicted quality or higher dissimilarity counts as more likely defective; every sign and primary score was fixed before scoring and no threshold was tuned. Dropping the split-vote scenes (Tab. S5) likely removes hard cases, and the clip judge's prompt counts any confirmed difference as wrong, stricter than annotators may have been, so label noise can move these scores either way. Raising the clip judge's own threshold trades recall for precision: the first threshold at which at least 70% of its flags are defective keeps 21 flags holding 15 of the 29 defective scenes, and the first at 85% keeps 8 holding 7. DOVER [64] and VideoScore2 [24] never see the catalog photos, so they cannot tell a wrong product from the right one; they show that a standard quality score would pass such scenes (VideoScore2 gives 107 of the 120 clips its top visual-quality score). The open judges received the deployed prompt, photos and output schema; Qwen3.5-9B [51] ran in

Table S2. The same ads with judges hidden (60 ads each). Alone: share of units the reviewer marked with every judge hidden. Judge: share the stored findings flagged. Alone's marks flagged: of the units the reviewer marked, the share the judge had flagged. Flags marked alone: of the judge's flags, the share the reviewer marked. Both columns give counts when fewer than 20. Round any and majority: share marked by any round reviewer, or by more than half of those asked, with judge findings on screen. κ: Cohen's κ between the reviewer alone and the round's majority on units both were asked. Shares in %, brackets are 95% intervals. R2 was also one of the round's reviewers on these ads (the only one on 20 of them), so the round columns include that reviewer's own review and their κ against the majority is not given.

Axis	Unit	Units	Alone	Judge	Alone's marks flagged	Flags marked alone	Round any	Round maj.	κ alone vs maj.
R1 (60 ads)									
Product	scene	280	33 [25, 41]	68	74 [61, 85]	36 [27, 44]	67	58	0.11 [−0.02, 0.22]
Scene quality†	scene	280 (229)	47 [38, 55]	25	27 [20, 33]	49 [37, 61]	37	33	0.05 [−0.07, 0.17]
Script	line	280	3 [1, 4]	3	0 of 7	0 of 7	15	5	−0.03 [−0.05, −0.02]
Caption placement	caption	255	13 [8, 19]	no judge			17	5	0.06 [−0.06, 0.21]
Caption relevance	caption	255	5 [2, 8]	no judge			6	2	−0.03 [−0.05, −0.01]
R2 (60 ads)									
Product	scene	280	19 [13, 26]	68	93 [85, 98]	26 [18, 34]	67	58	
Scene quality†	scene	280 (229)	31 [26, 36]	25	41 [31, 51]	49 [37, 62]	37	33	
Script	line	280	5 [3, 8]	3	1 of 15	1 of 7	15	5	
Caption placement	caption	255	16 [10, 21]	no judge			17	5	
Caption relevance	caption	255	1 [0, 3]	no judge			6	2	

bf16 and Qwen3.6-27B [52] in FP8, with thinking on as the primary setting (sampled at the temperature their model cards recommend, one sample per scene; thinking off was decoded greedily). A repeat with another seed and server layout moved the 9B's AUC by up to 0.09, so single rows carry that much run-to-run noise beyond the interval shown. The best open setting, the 27B with thinking off on eight frames, stays 0.12 below the clip judge, with an interval that includes zero, the only such gap besides the deployed judge's own eight-frame run; its verdicts reach κ 0.35 (recall 17 of 29, precision 17 of 36), against 0.43 (26 of 29, 26 of 56) for the clip judge. With thinking on, the open models flag 69 to 99 of the 120 scenes (κ 0.05 to 0.15). The prompt was written for the closed judge, and no variant was tried for the open ones. The clip judge's earlier runs and the starting-frame judge did not record their model version, so the eight-frame and starting-frame gaps may include a model change; the eight-frame run keeps the deployed prompt, while the starting-frame judge has its own (Tab. 4).

Table S3. Product-fidelity baselines on the 120 blind-labeled scenes. Gap: clip judge minus baseline, paired, with 95% interval; intervals are not corrected for multiple comparisons.

Score	Input	AUC	95% interval	Gap to clip judge
Clip judge (deployed)	clip, catalog photos	0.84	[0.73, 0.94]	
Starting-frame judge	starting frame, photos	0.69	[0.61, 0.78]	+0.15 [0.01, 0.28]
Clip judge on eight frames	8 frames, photos	0.80	[0.66, 0.92]	+0.04 [−0.05, 0.15]
DINOv2 similarity	3 frames, photos	0.51	[0.39, 0.65]	+0.34 [0.18, 0.48]
CLIP similarity	3 frames, photos	0.57	[0.41, 0.72]	+0.27 [0.09, 0.47]
OCR text mismatch (EasyOCR)	3 frames, photos	0.57	[0.49, 0.64]	+0.28 [0.14, 0.42]
OCR text mismatch (Tesseract, secondary)	3 frames, photos	0.58	[0.50, 0.64]	+0.27 [0.14, 0.39]
DOVER (overall)	clip, no photos	0.48	[0.34, 0.63]	+0.37 [0.19, 0.56]
VideoScore2 (mean of 3 scores)	clip, text prompt	0.57	[0.44, 0.69]	+0.28 [0.12, 0.44]
Qwen3.5-9B, thinking on	8 frames, photos	0.67	[0.49, 0.84]	+0.18 [0.04, 0.31]
Qwen3.5-9B, thinking on	clip, photos	0.69	[0.55, 0.82]	+0.15 [0.04, 0.26]
Qwen3.6-27B (FP8), thinking on	8 frames, photos	0.72	[0.59, 0.86]	+0.12 [0.03, 0.22]
Qwen3.6-27B (FP8), thinking on	clip, photos	0.67	[0.53, 0.81]	+0.18 [0.06, 0.29]
Qwen3.6-27B (FP8), thinking off	8 frames, photos	0.73	[0.60, 0.86]	+0.12 [−0.04, 0.26]
Qwen3.6-27B (FP8), thinking off	clip, photos	0.68	[0.57, 0.80]	+0.16 [0.01, 0.29]
Qwen3.5-9B, thinking off	8 frames, photos	0.54	[0.39, 0.71]	+0.30 [0.15, 0.46]
Qwen3.5-9B, thinking off	clip, photos	0.60	[0.48, 0.74]	+0.25 [0.10, 0.38]

D. Label Sets Behind the Numbers

Table S4 lists two more experiments of Sec. 6, with their labels and answers. Table S5 lists each label set behind the judge, review and fix numbers.

Table S4. Other judge experiments. Labels: **B** listeners blind to production history, at least two per voice, **L** one listener per line; numbers measure agreement with that reference only. Each choice was made and scored on the same labels, so numbers are optimistic.

Axis and question	Labels	Answer and key numbers
Pronunciation. Which speech-to-text model; recheck on audio?	**L** 208 lines: audio only (blind), then with script and judge output	**Error-keeping, text-only recheck.** On the labels made after output was shown (54 defect lines, against 43 on the blind pass of Tab. 2), Gemini 3.5 Transcribe [44] and ElevenLabs Scribe v2 [14] returned the script word for word on 30% and 46% of defect lines, against 6% for ours, Azure Speech [46]; MAI-Transcribe-2 [47] also caught fewer defects than ours. Audio in the recheck cut recall from 44 to 38 of 54
Sound. Which text-to-speech voices to allow?	**B** 107 voices on an earlier engine, three ad reads each; four listeners	**65 of 107,** excluding voices with poor recording quality, mispronunciations, unnatural intonation or an accent that did not fit the target market. Each voice was heard by at least two listeners, and the 44 split voices went to a tiebreak with earlier votes hidden; two voices users had swapped away from in production were among the 42 excluded, and the three most-cast voices were kept. The engine that narrated the round was admitted with all 30 of its voices

E. Models

The judges whose findings reviewers saw in the round run on closed models: Gemini Flash for the product clip judge, the scene clip judge, the sound judge and the listing-photo text reader; a GPT-5.6 model for the scene frames judge; Azure Speech with a GPT-5.6 model for pronunciation; and Claude Sonnet for the script judge. The caption judges, run only after the round, use Gemini Flash on the whole rendered ad. Gemini names are aliases that the provider updates, so a rerun may reach a newer model.

The edit techniques of Tab. 3 can run on several kinds of models. Repair needs a generative upscaler that regenerates the scene's own clip at higher resolution (video-to-video), such as SeedVR2 [61], STAR [66] or Topaz Starlight [60]. A new scene needs a reference-to-video model, and a clip edit a video-editing model such as Gemini Omni, Seedance or Kling. Lines can be voiced with text-to-speech voices such as Gemini's or ElevenLabs', and a whole ad re-voiced speech-to-speech. Line suggestions can come from a language model, and music from a text-to-music model such as Lyria or Eleven Music. A logo is cut out from its background by a background-removal model. At generation, the quoted-line rewrite pairs the script judge with a language-model writer. Moving or restyling a caption uses a vision-language model and rewording one a language model (Tab. 3); borrowing a scene and moving a logo call no model.

F. Review Rubric

Table S6 lists the questions of the review rubric as they were used in the 120-ad round (Sec. 4). The scene-quality grid asked for one answer on every rateable scene, meaning a scene with a generated clip or an animated still and never the end card; for most of the round the grid skipped scenes already marked product wrong and counted them as swaps.

Table S5. Label sets. "Blind" means the labelers saw no judge output. Assisted labels were made with judge findings on screen, and a thumbs up wrote the finding into the answer.

Label set	Size	Labelers	Blind?	Used for
Annotator product labels on production clips	120 decided scenes of 146 labeled, 29 defective	four annotator accounts; of the 120 decided scenes, 105 have two votes, ten have three or four, five have one; they miss subtle defects	yes	product judge and baselines (Tab. 4, Tab. S3)
Earlier review round	66 accepted scene findings	reviewer thumbs	assisted	what scene judges flag
Scene judges, 120-ad round	489 reviewer-scene pairs	portal reviewers	assisted	one scene judge or two (Tab. 4)
Pronunciation corpus	208 narrated lines, earlier narration engine	five listeners, one per line, ten lines heard by all	first pass yes, second pass after reveal	recall and precision on the first pass (Tab. 2); speech-to-text model and recheck on the second (Tab. S4)
Fresh pronunciation sample	100 production lines (70 flagged, 20 near misses, 10 passed)	one listener	yes	precision on the 70 flagged lines: 0.21 [0.12, 0.31]
Talking-over clips	50 of 128 clips	stem-separation loudness	no human labels	sound (Tab. S4)
Script judge study	155 ads, 200 ratings	four raters, shown both judge versions' calls	no	reading listing photos
Caption placement	100 captions	models against themselves	no human labels	placement pick stability (Tab. 4)
Caption marks, 120-ad round	530 captions	portal reviewers	yes, no caption judge shown	caption judges, run after the round; the placement prompt was chosen on these marks (Tabs. 2 and 4)
120-ad review round	562 scenes, 278 reviews	five internal reviewers	assisted	split, time, agreement (Sec. 4)
Reviewers alone	60 of the round's ads, 280 scenes	two reviewers, R1 and R2, 60 ads each	R1 yes; R2 had seen judge findings in the round	the same ads with judges hidden (Supp. B)
Repair test	30 scenes with garbled product text, 26 rated	one internal reviewer	yes	Repair (Sec. 5)
Respelling test	75 flagged lines, two takes each	pronunciation judge only	no human labels	respelling against a plain second take (Supp. A)
Quoted-line rewrite	129 scripts the judge cleared after the rewrite, of 134 it flagged in 410 earlier ads	two annotators, then an adjudicator; no controls	first verdict yes	regenerating only the quoted lines (Sec. 5)
Voice listening study	107 voices on an earlier narration engine, three ad reads each	four listeners, at least two per voice; split voices to a tiebreak	yes (blind to production history)	voices to allow (Tab. S4)

Table S6. Questions of the review rubric in the 120-ad round, in the order the portal showed them in the round. The last answer of each question is the problem answer; it opens the picker and the secondary options. The two overlay-text rows are the two parts of the captions axis of Tab. 1.

Question	**Answers**	**A problem answer points to**	**Secondary options**
The product	Product looks right; Product wrong	every scene where it happens	Wrong product; Shape, color, logo or markings wrong; AI issue; Product text not intact, graded Low ("still readable") or High ("gibberish"); a comment per scene
Scene quality	No opinion; Keep; Swap out	each rateable scene, which gets its own answer	AI issue; Repetitive, dull or crowded scene; World or continuity breaks; Something moves impossibly; Weak ending (last scene only); free-text boxes for another reason and for why
Pronunciation	Sounds right; Mispronounced	every spoken line where it happens	none; an optional box for the word
Sound quality	Sounds right; Issues	the whole ad	"Which part?", at least one of The voice and The music; an optional note
Script quality	Good; Bad	every spoken line where it happens	Factually incorrect (an unsupported claim); Awkward or unnatural phrasing; an optional note
Overlay text: placement	Correct; Can be better	every captioned scene where it happens	none
Overlay text: relevance	Correct; Issues	every captioned scene where it happens	none